Becoming-Sound

Affect and Assemblage in Improvisational Digital Music Making

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Outline

- Affect and assemblage
- Jamming in *Viscotheque*: a case study

The *intensity* of

jamming

complex socio-technical
interactive systems

beyond usability...

...towards *experience*

experience

beautiful snowflake

all the same

affect

affect / intensity

Spinoza

Deleuze

Massumi

affect	pre-personal
feeling	personal, biographical
emotion	performed, projected

affect is *unstructured potential*

feelings represent the *registration* of affect

affect *gives rise to* movement, transformation and difference

what can a body do?

assemblage is the network(ing) of affective/intensive forces

...which can open up new potentials for action agency is complex

posture

toe-tappin'

thigh-slappin'

music is *affective*

mood

memory

focus

manipulative

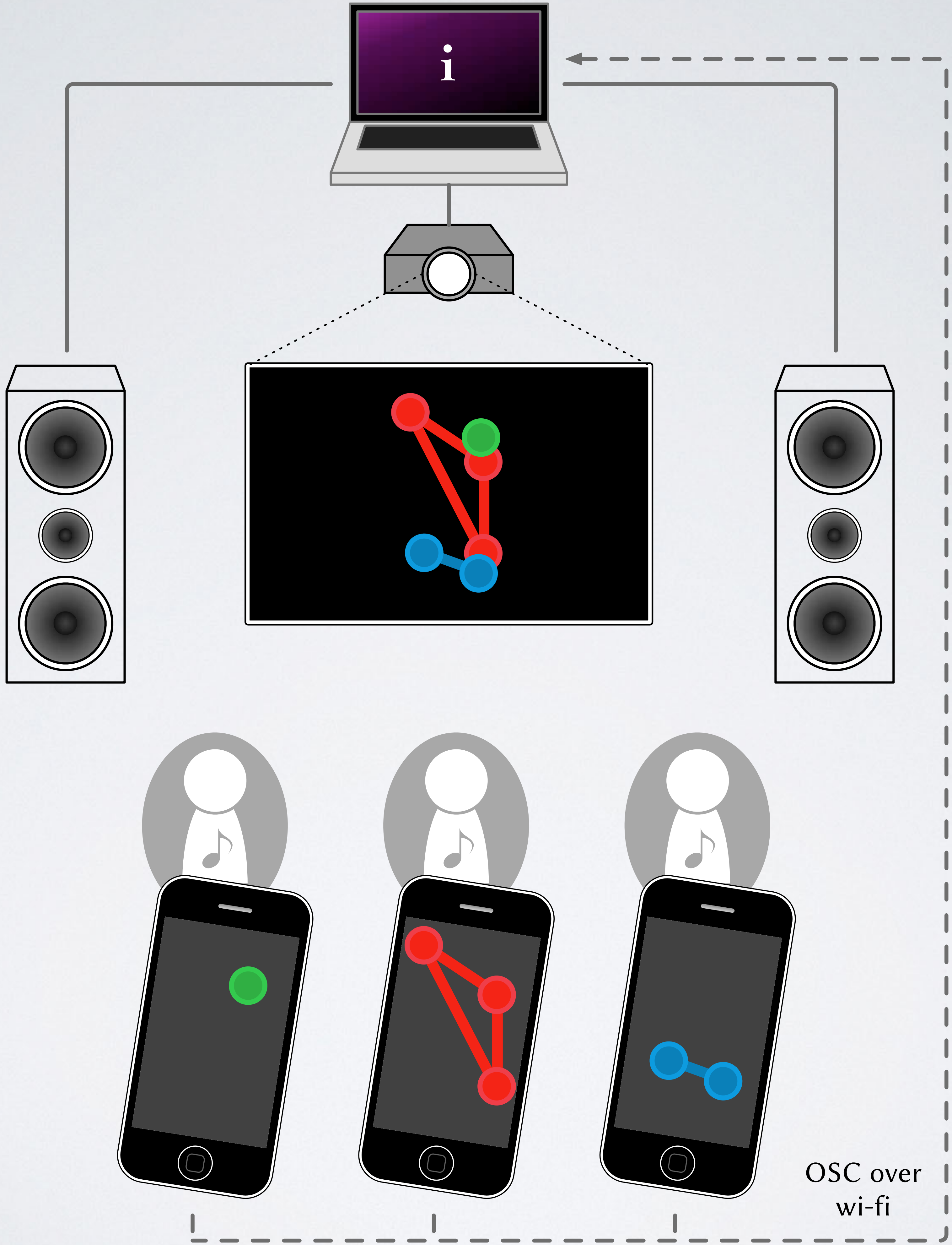
concentration

Viscotheque

iOS group

music making

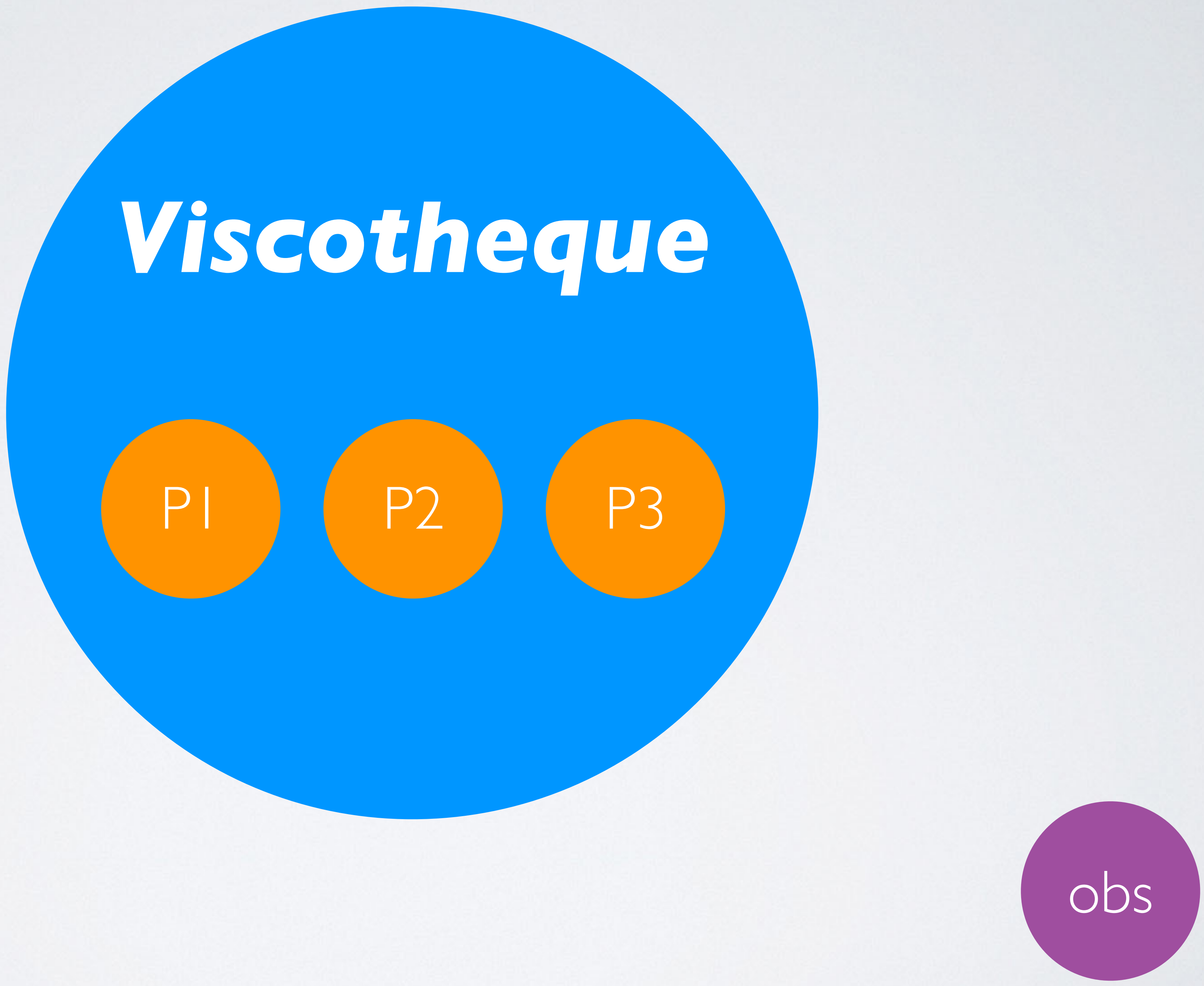
environment

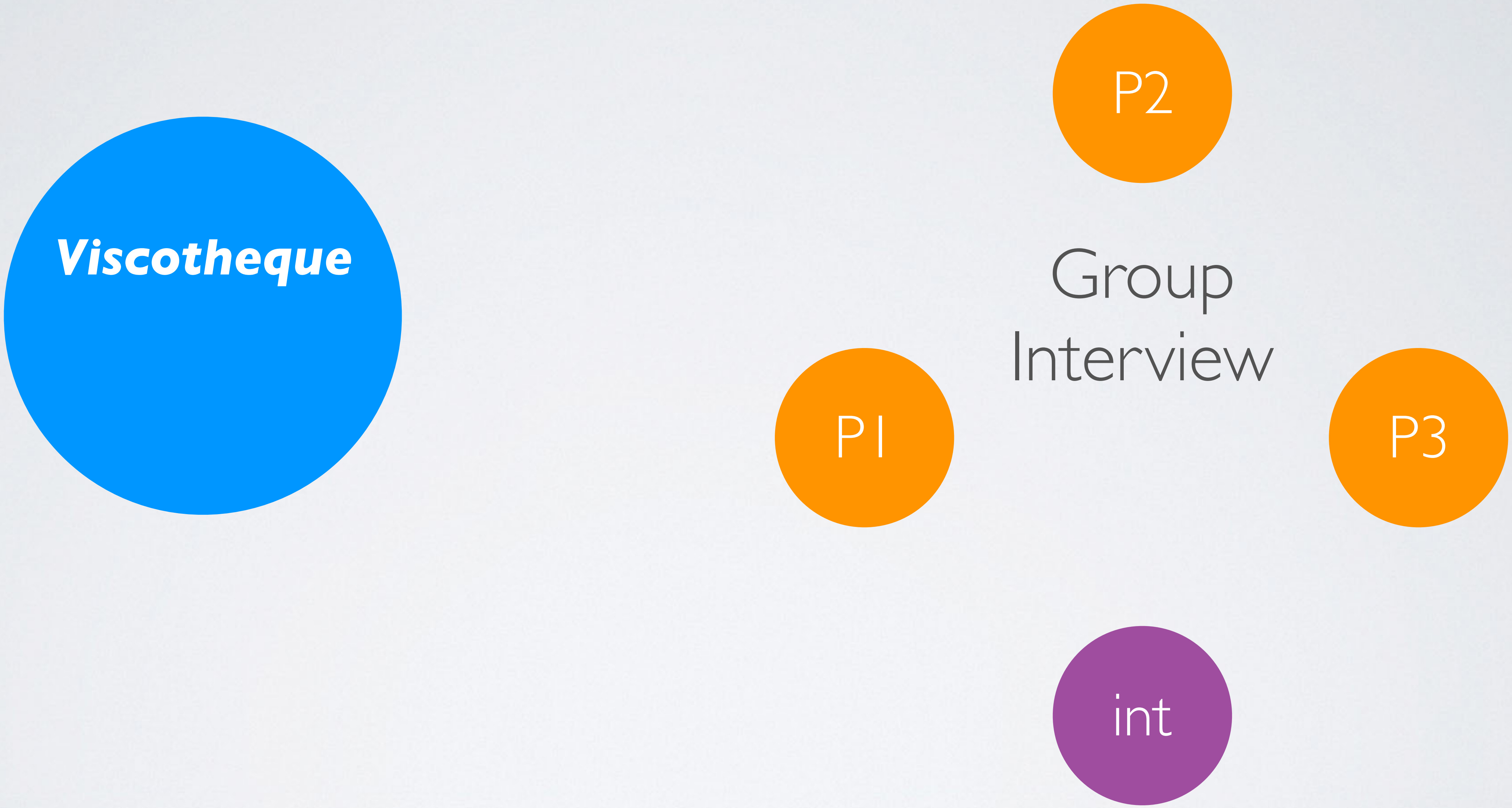


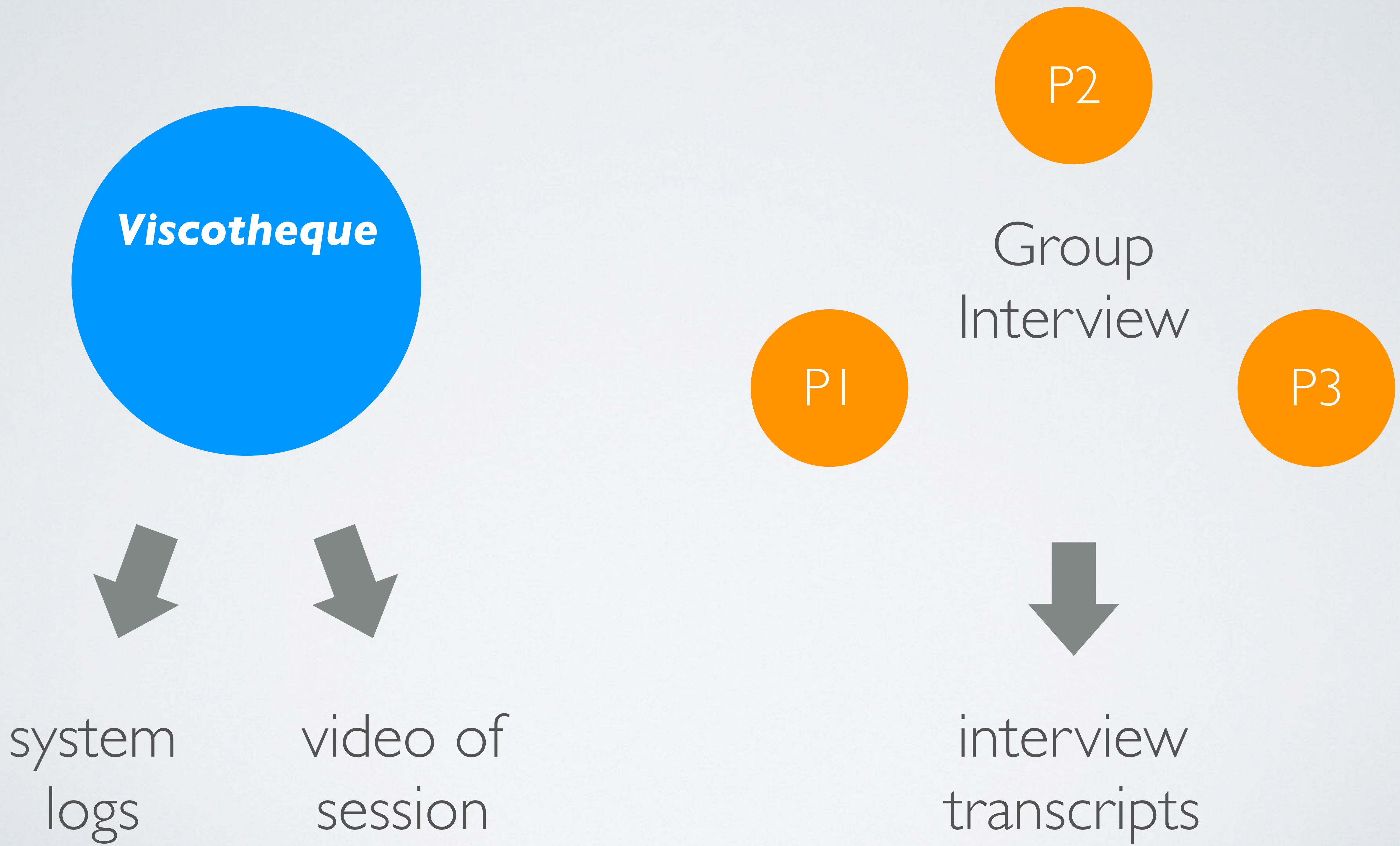
i
OSC over
wi-fi

what are the affective atmospheres in Viscotheque?

Viscotheque
P1
P2
P3
Participants
obs

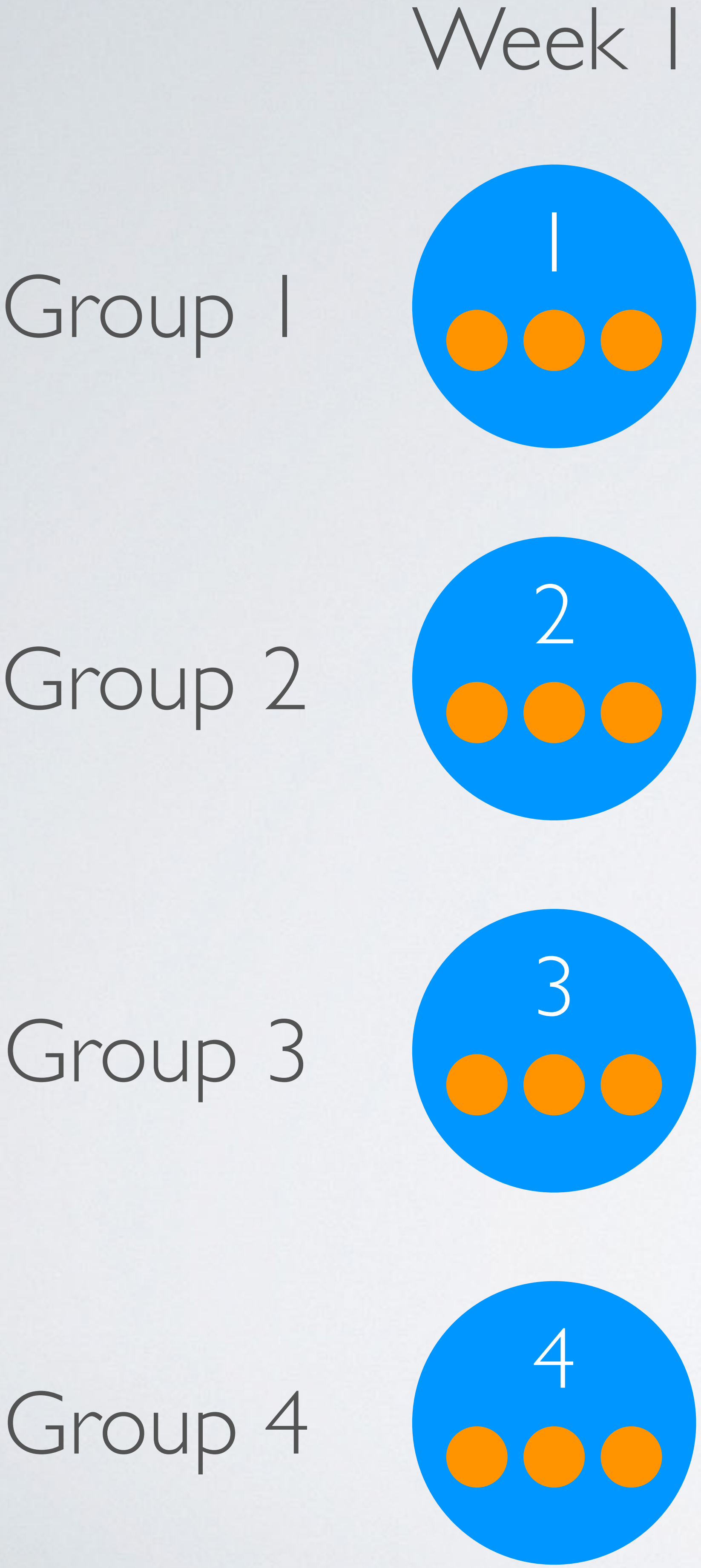

Viscotheque
P1
P2
P3
obs

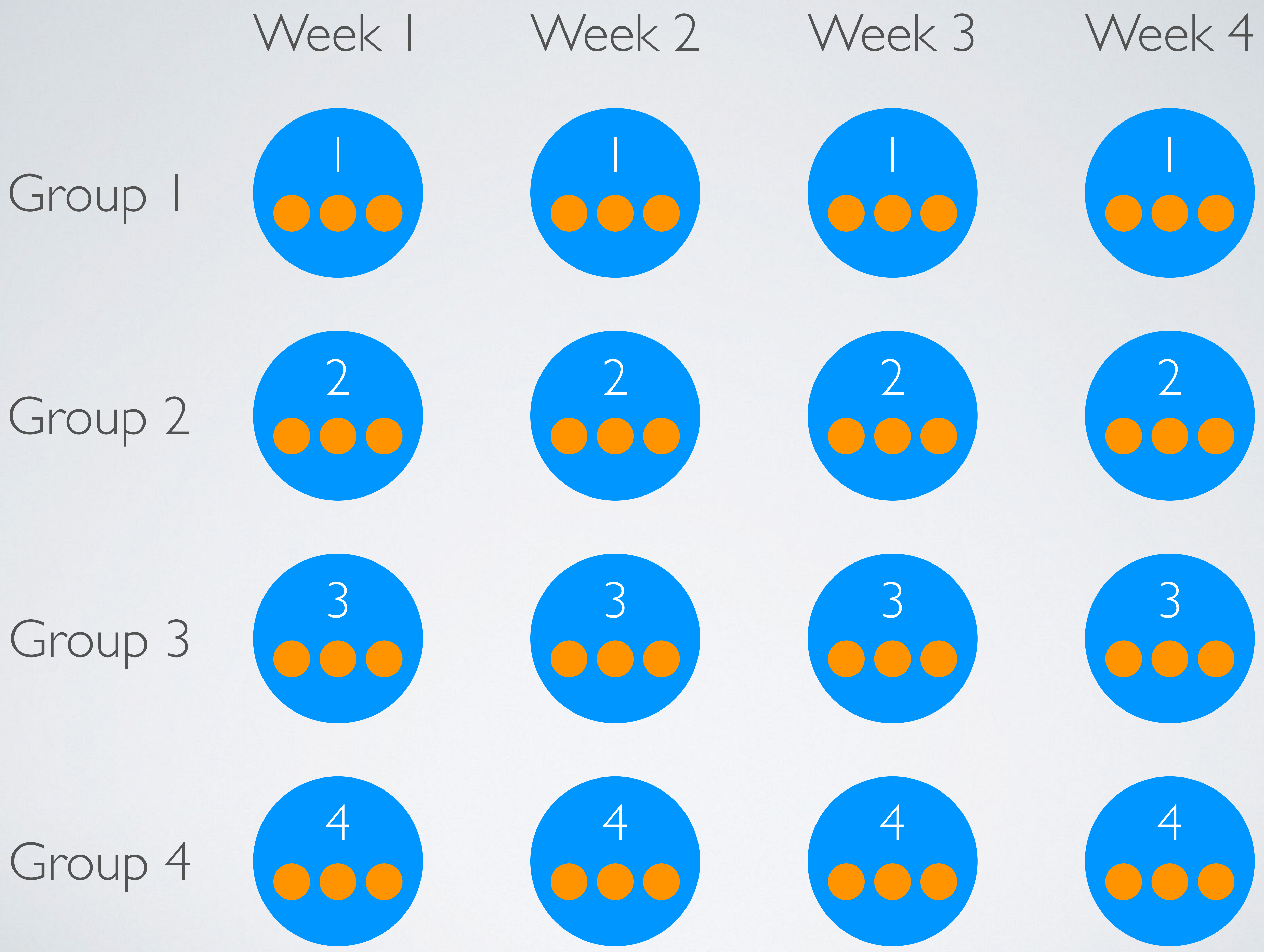

Viscotheque
P2
Group
Interview
P1
P3
int


Viscotheque
P2
Group Interview
P1
P3
system logs
video of session
interview transcripts

Week 1

Group 1


Week 1
Group 1
1
Group 2
2
Group 3
3
Group 4
4


Week 1
Week 2
Week 3
Week 4
Group 1
Group 2
Group 3
Group 4
1
2
3
4

Interview themes

- 50k words, grounded theory
- The affective power of the sound
- Ownership of/identification with the sound
- The need to differentiate, to *find* a sound

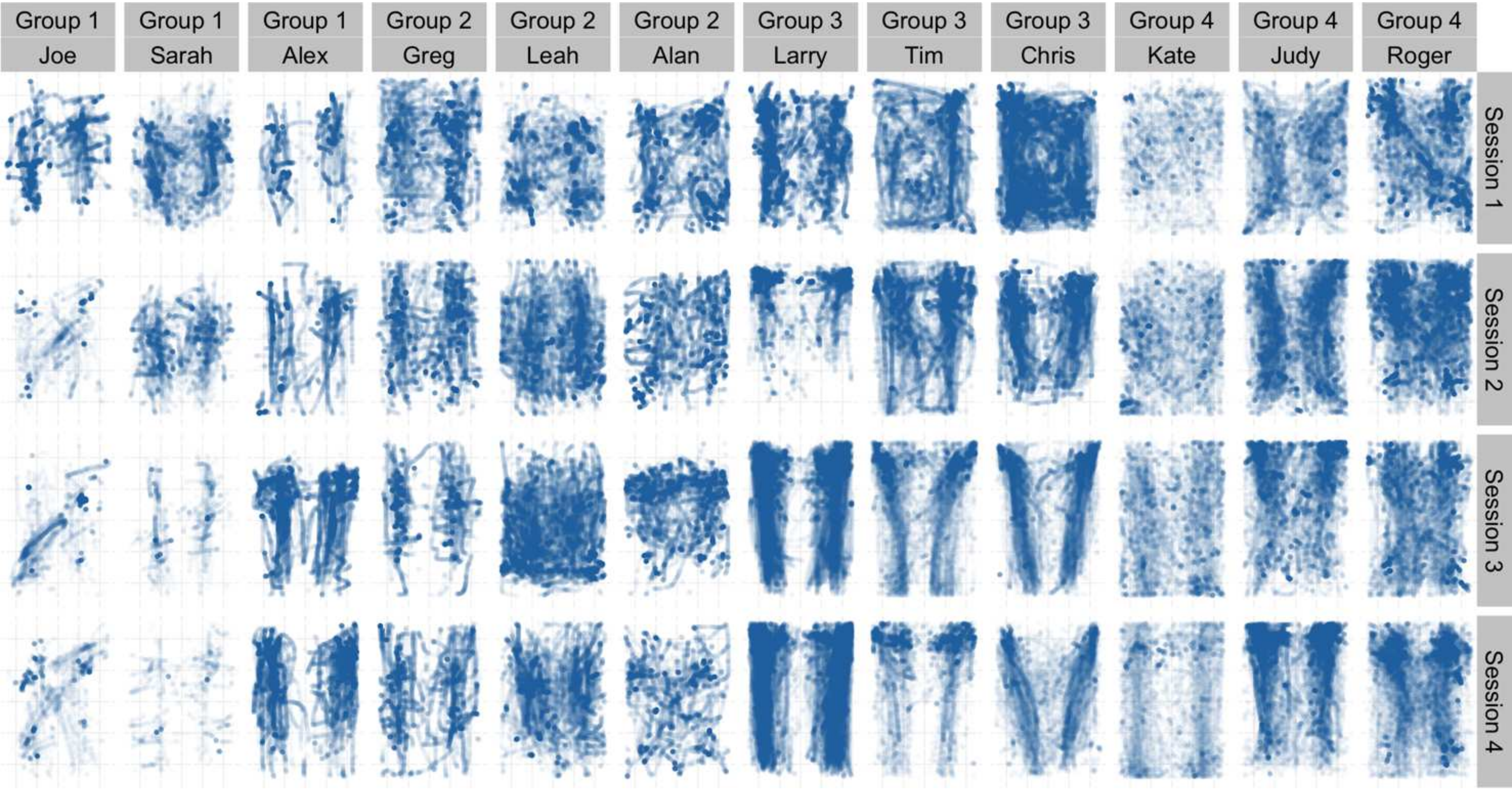
Observations

- Moments of real intensity
- Earlier sessions—flighty, skittish
- Later sessions—sedate, subdued
- Punctuated equilibrium

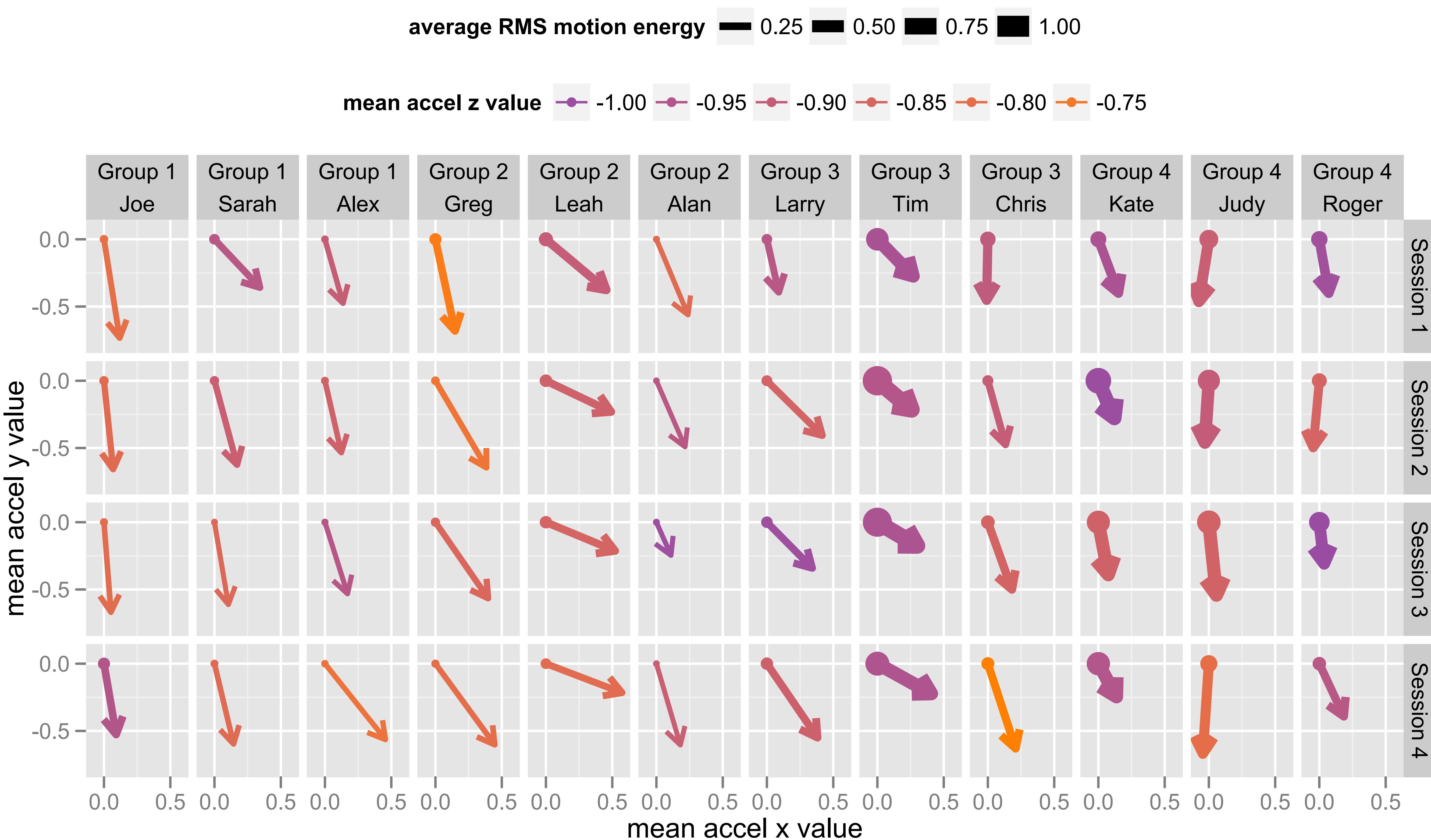
becoming-sound

but we have all this log data...

accelerometer signatures



Group 1
Joe
Group 1
Sarah
Group 1
Alex
Group 2
Greg
Group 2
Leah
Group 2
Alan
Group 3
Larry
Group 3
Tim
Group 3
Chris
Group 4
Kate
Group 4
Judy
Group 4
Roger
Session 1
Session 2
Session 3
Session 4


average RMS motion energy
0.25
0.50
0.75
1.00
mean accel z value
-1.00
-0.95
-0.90
-0.85
-0.80
-0.75
Group 1 Joe
Group 1 Sarah
Group 1 Alex
Group 2 Greg
Group 2 Leah
Group 2 Alan
Group 3 Larry
Group 3 Tim
Group 3 Chris
Group 4 Kate
Group 4 Judy
Group 4 Roger
Session 1
Session 2
Session 3
Session 4
0.0
-0.5
0.0
0.5
mean accel y value
mean accel x value

Implications

- Agency is complicated in affectively-charged environments

Future work

- Co-location vs remote interaction
- Instrumentation: GSR, motion analysis

Cheers

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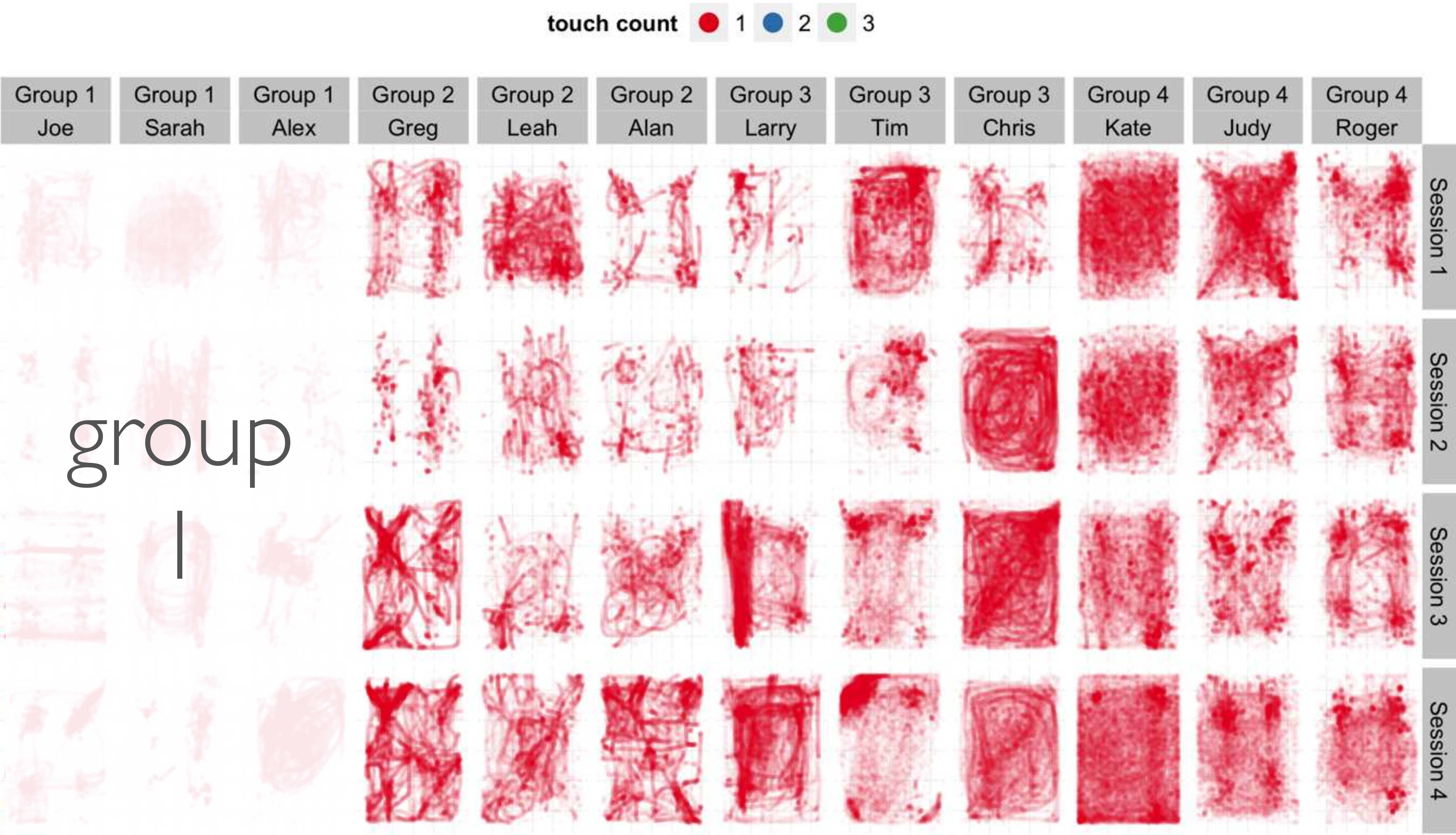
touch count
1
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Group 1
Joe
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Group 2
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Group 3
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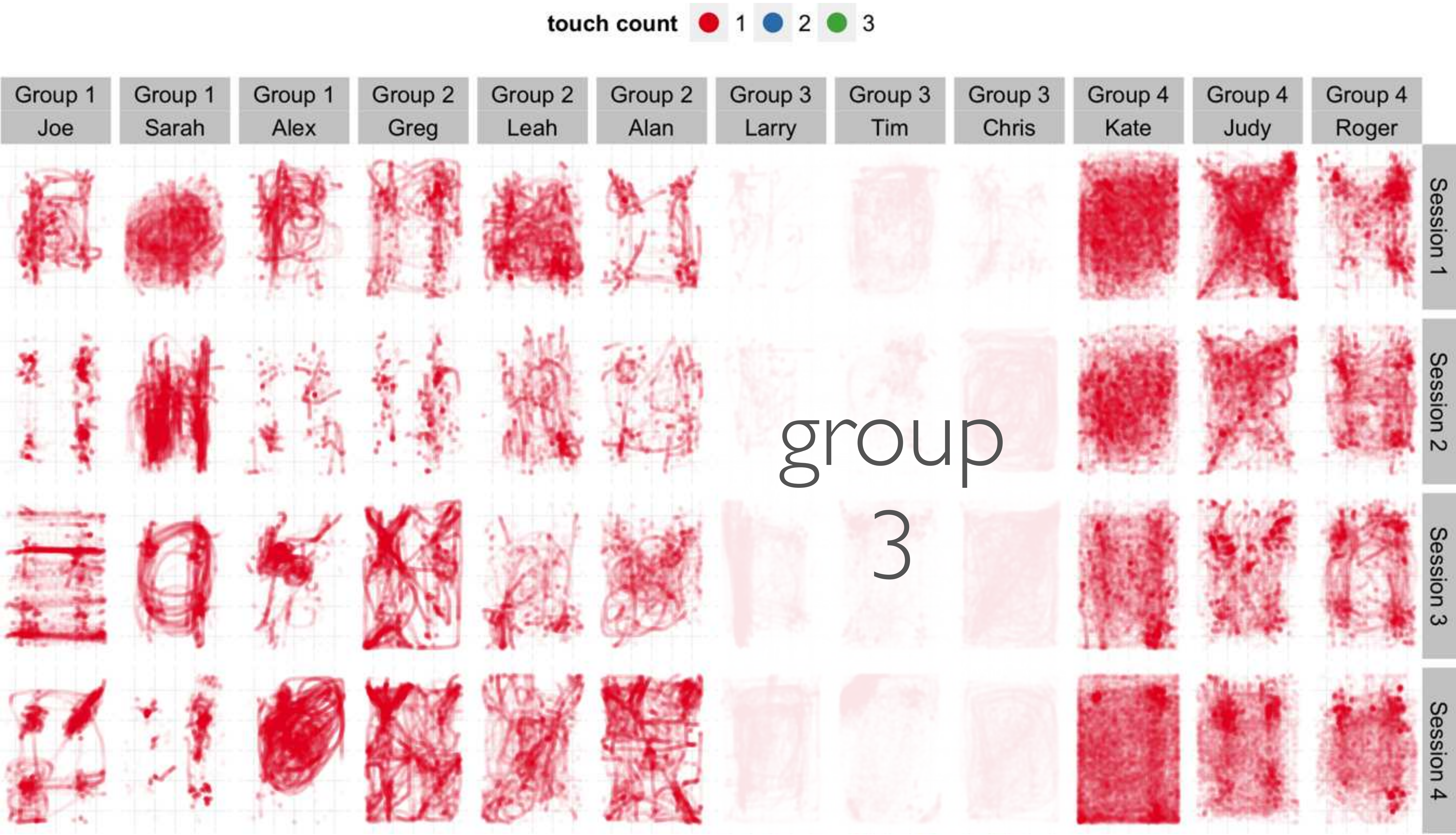
Group 3
Tim
Group 3
Chris
Group 4
Kate
Group 4
Judy
Group 4
Roger
Session 1
Session 2
Session 3
Session 4
group
1

touch count
1
2
3
Group 1 Joe
Group 1 Sarah
Group 1 Alex
Group 2 Greg
Group 2 Leah
Group 2 Alan
Group 3 Larry
Group 3 Tim
Group 3 Chris
Group 4 Kate
Group 4 Judy
Group 4 Roger
Session 1
Session 2
Session 3
Session 4
group 2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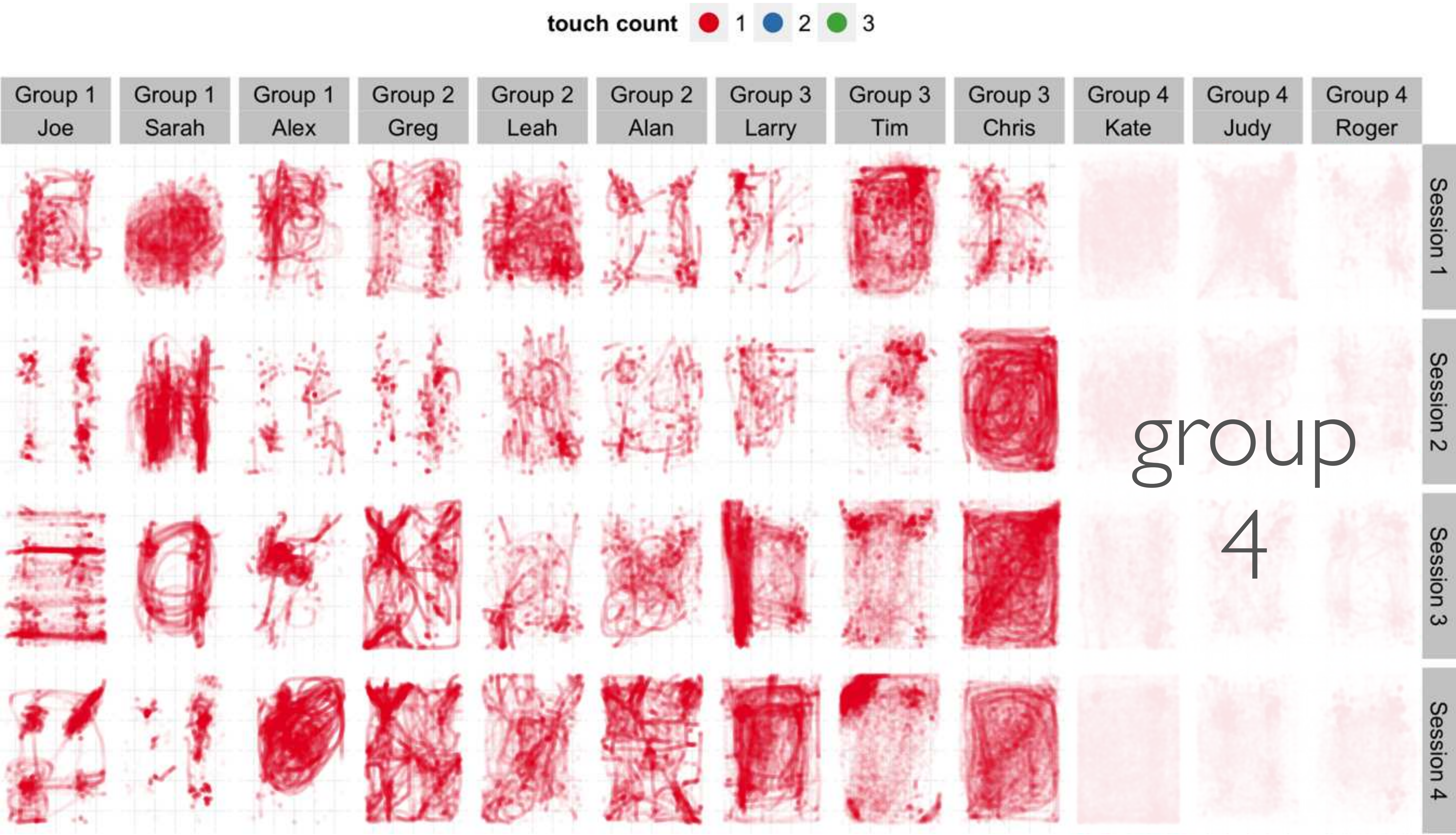
Group 4
Kate
Group 4
Judy
Group 4
Roger
Session 1
Session 2
Session 3
Session 4
group
3



touch count 1 2 3
Group 1 Joe
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Group 2 Leah
Group 2 Alan
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Group 3 Chris
Group 4 Kate
Group 4 Judy
Group 4 Roger
Session 1
Session 2
Session 3
Session 4
group 4


touch count
1
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3
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Group 1 Alex
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Group 2 Leah
Group 2 Alan
Group 3 Larry
Group 3 Tim
Group 3 Chris
Group 4 Kate
Group 4 Judy
Group 4 Roger
Session 1
Session 2
Session 3
Session 4
week 1


touch count
1
2
3
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Group 4 Roger
Session 1
Session 2
Session 3
Session 4
week 2

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Group 3 Tim
Group 3 Chris
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Group 4 Judy
Group 4 Roger
Session 1
Session 2
Session 3
Session 4
week 3


touch count
1
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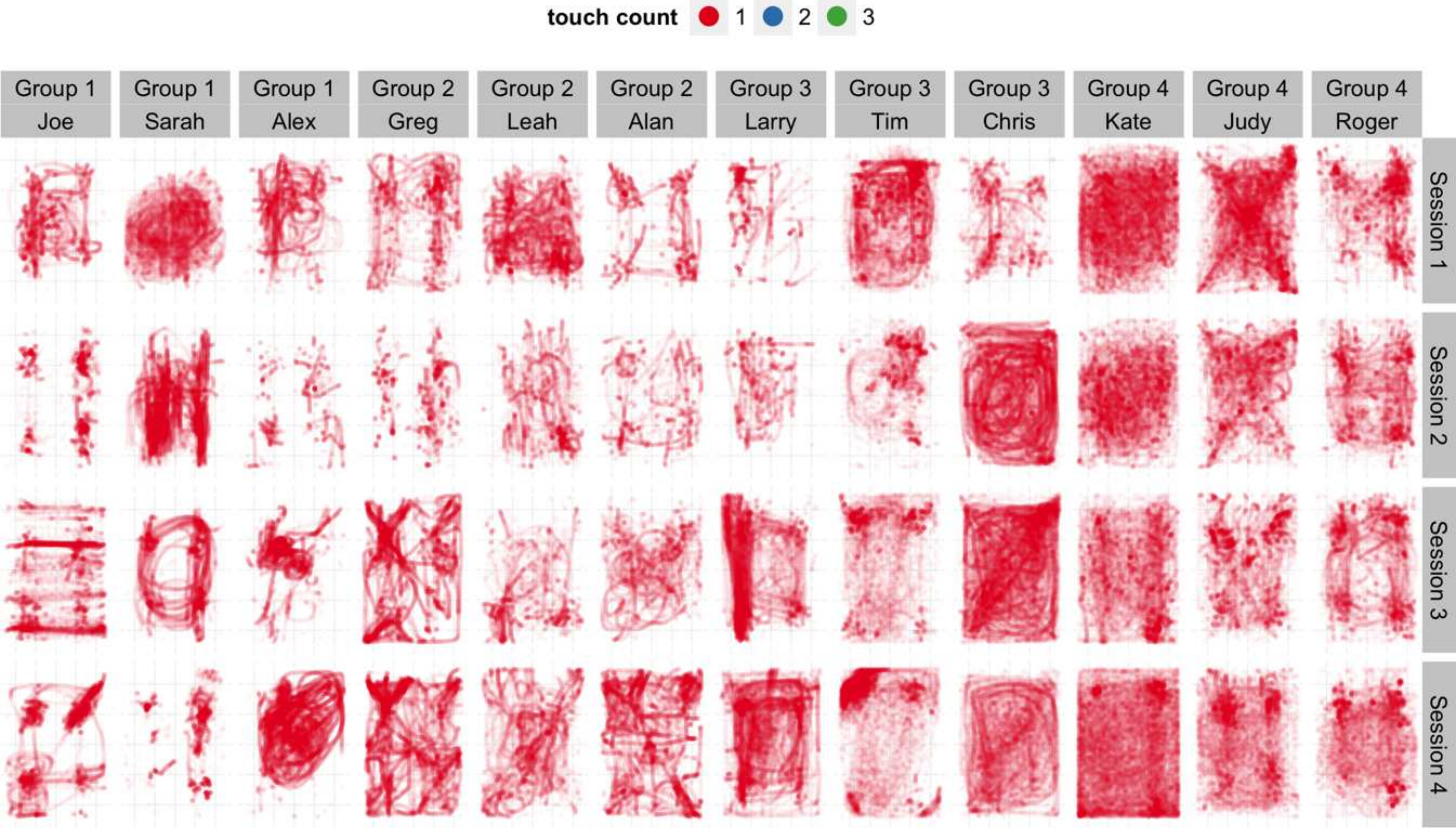
Group 4 Roger
Session 1
Session 2
Session 3
Session 4

week 4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Session 1
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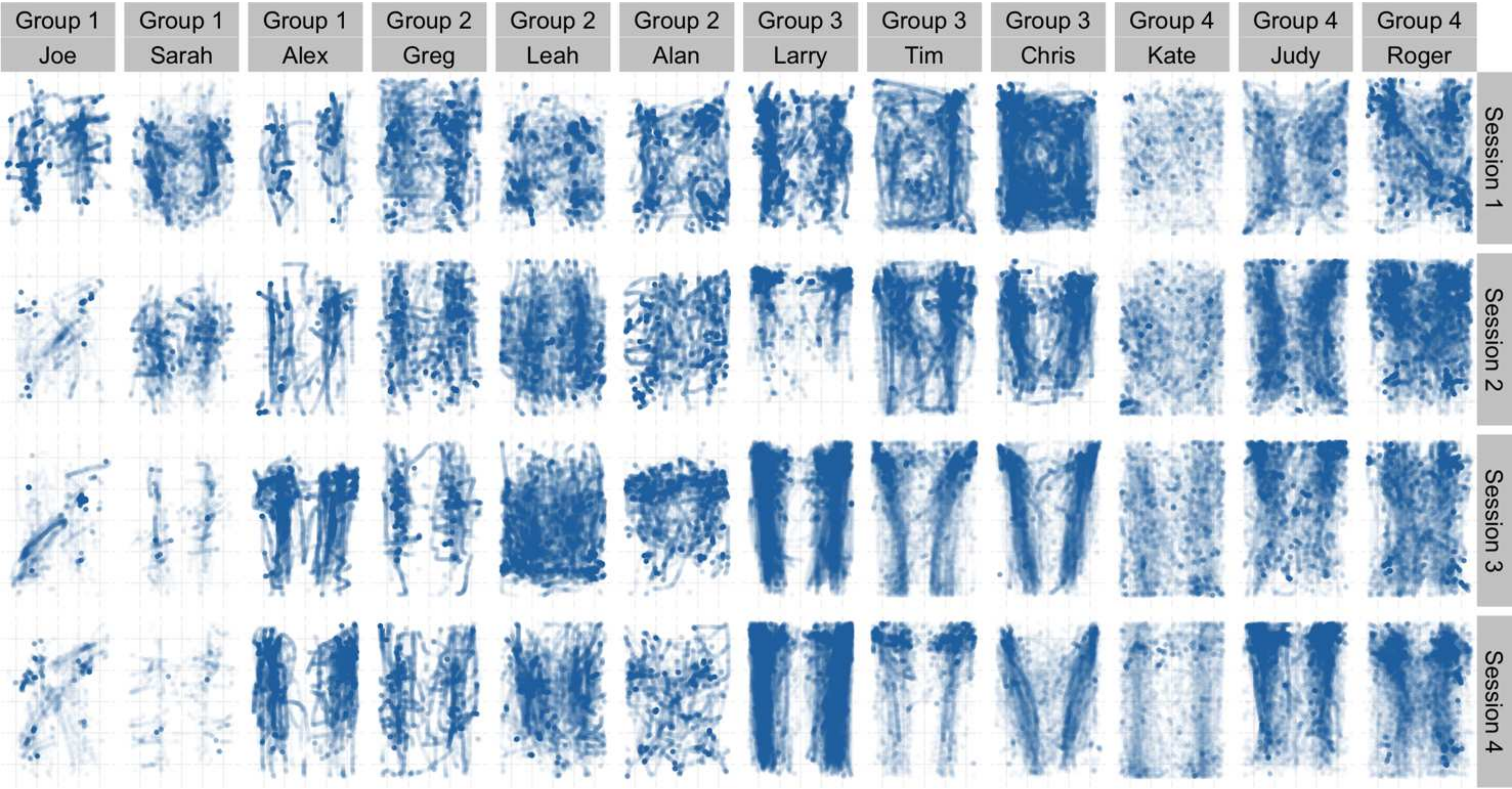
Session 4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Group 4
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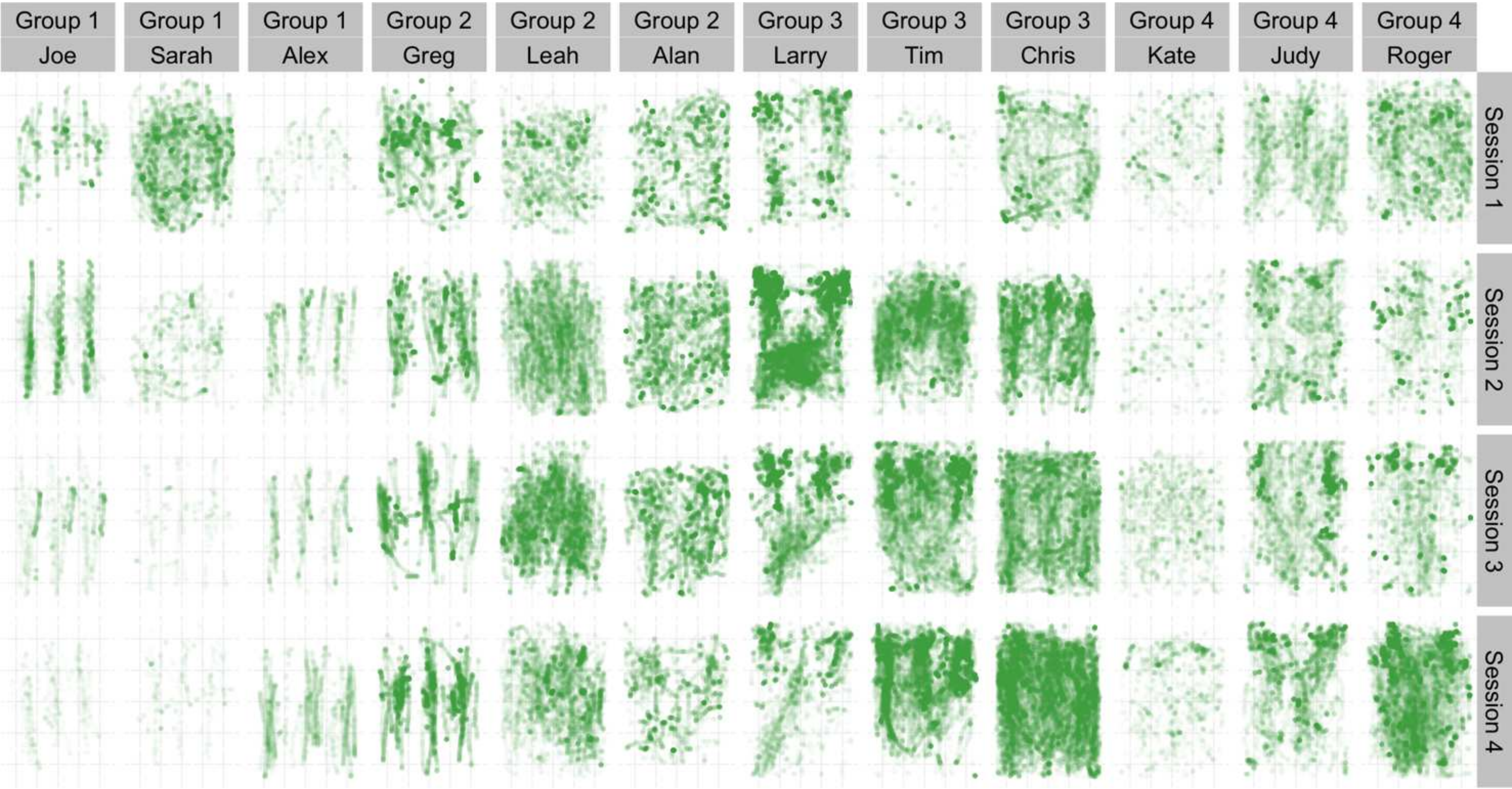
Session 1
Session 2
Session 3
Session 4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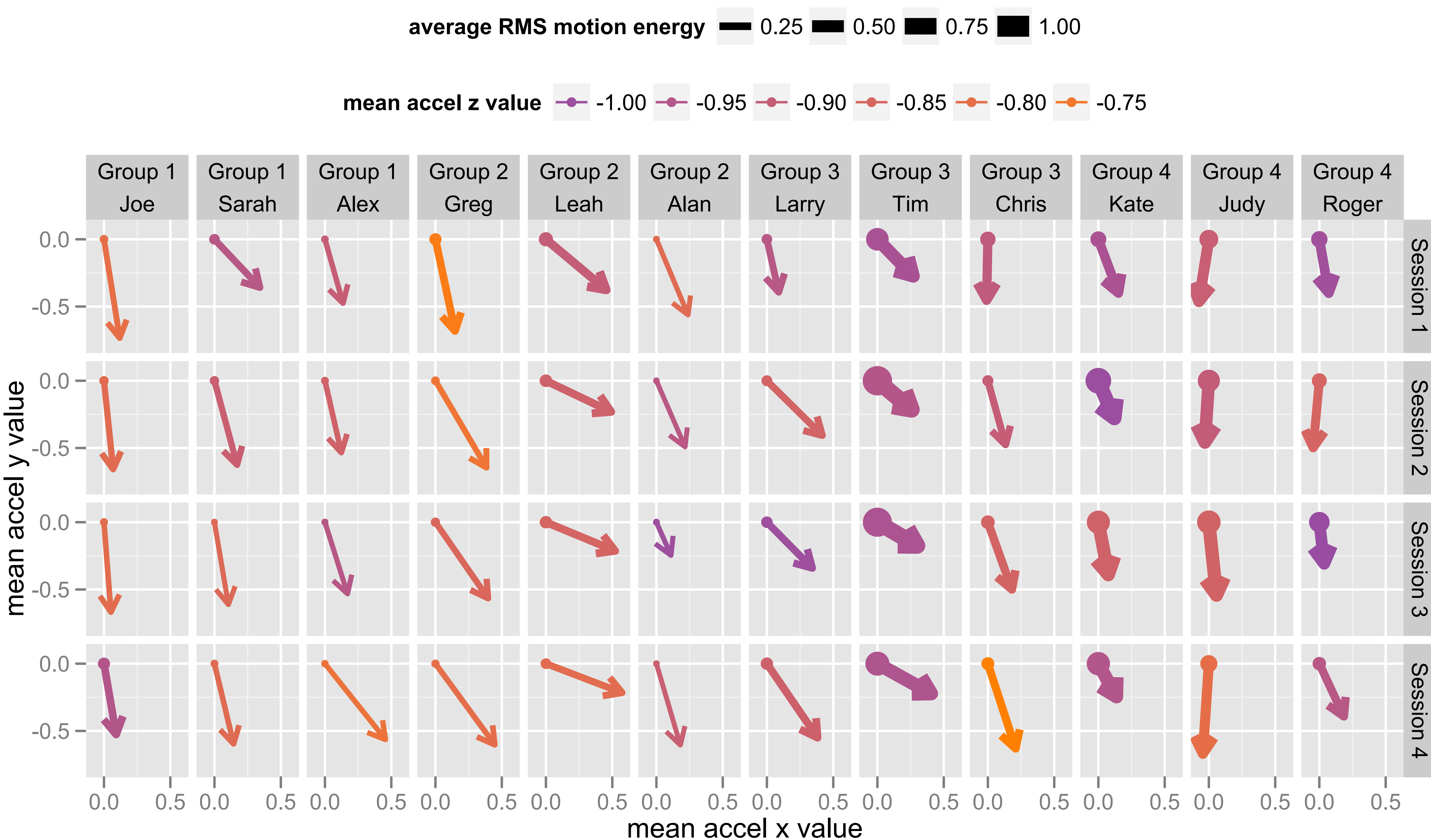
Session 1
Session 2
Session 3
Session 4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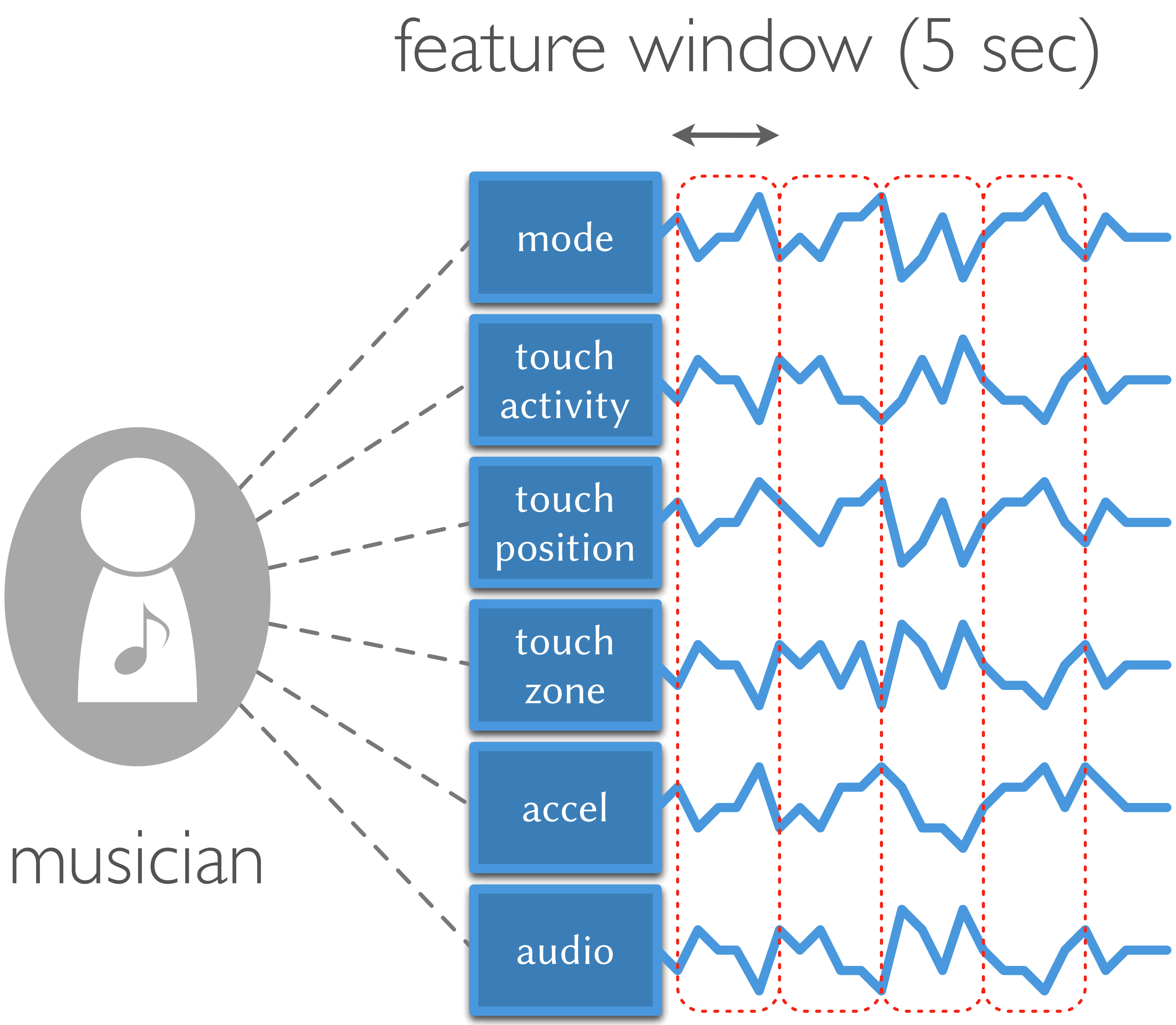
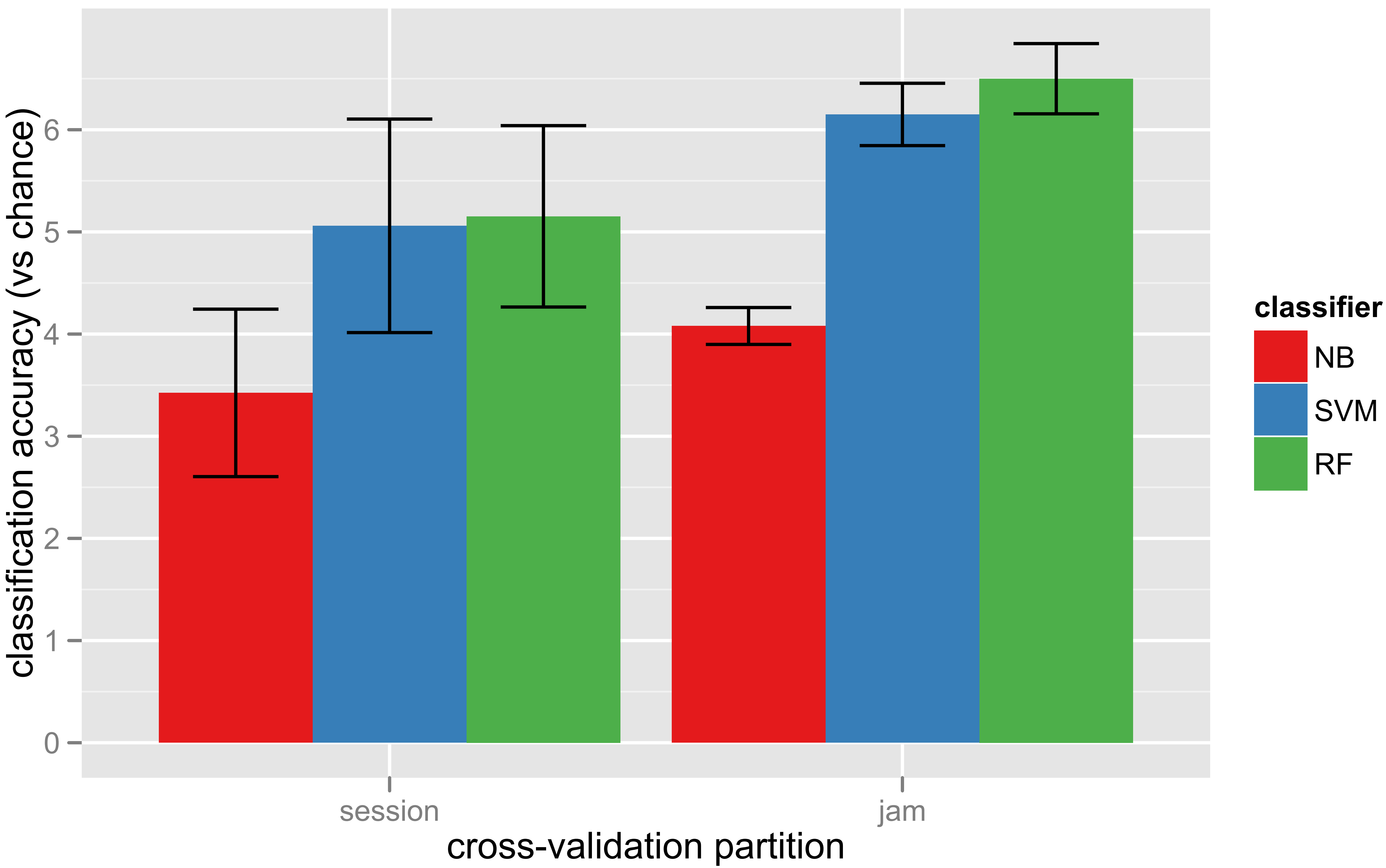
Group 3 Tim
Group 3 Chris
Group 4 Kate
Group 4 Judy
Group 4 Roger
Session 1
Session 2
Session 3
Session 4
0.0
-0.5
0.5
mean accel y value
mean accel x value

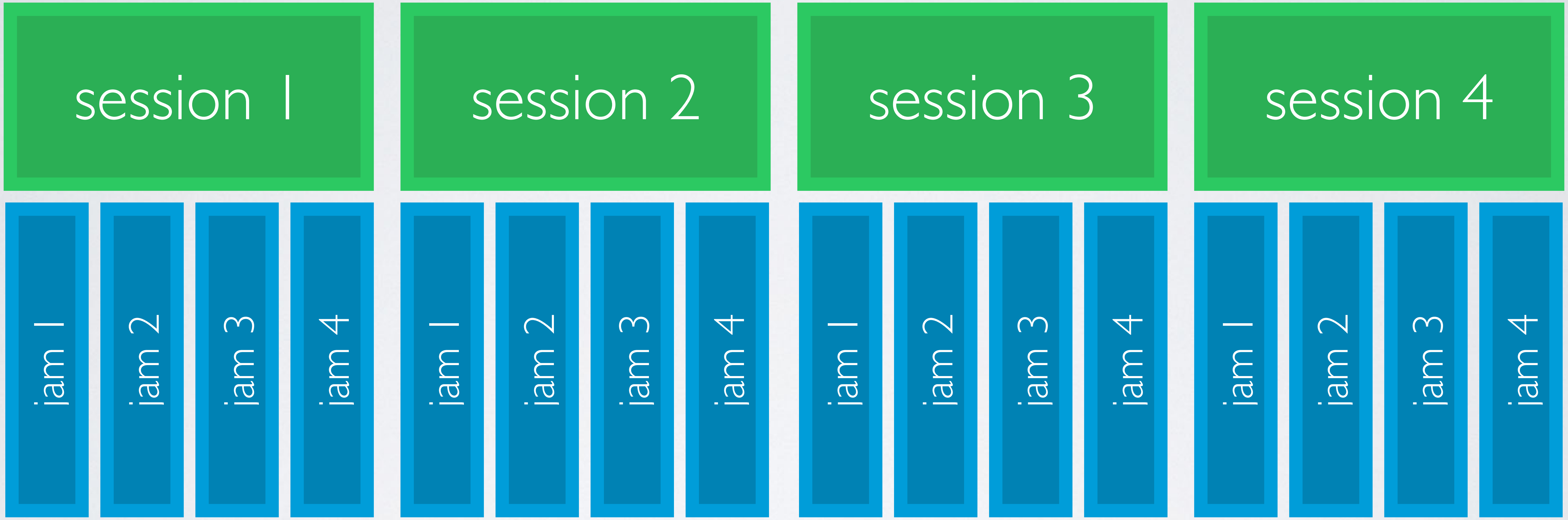
A *sensitive* synthesis of machine learning and affect theory requires...

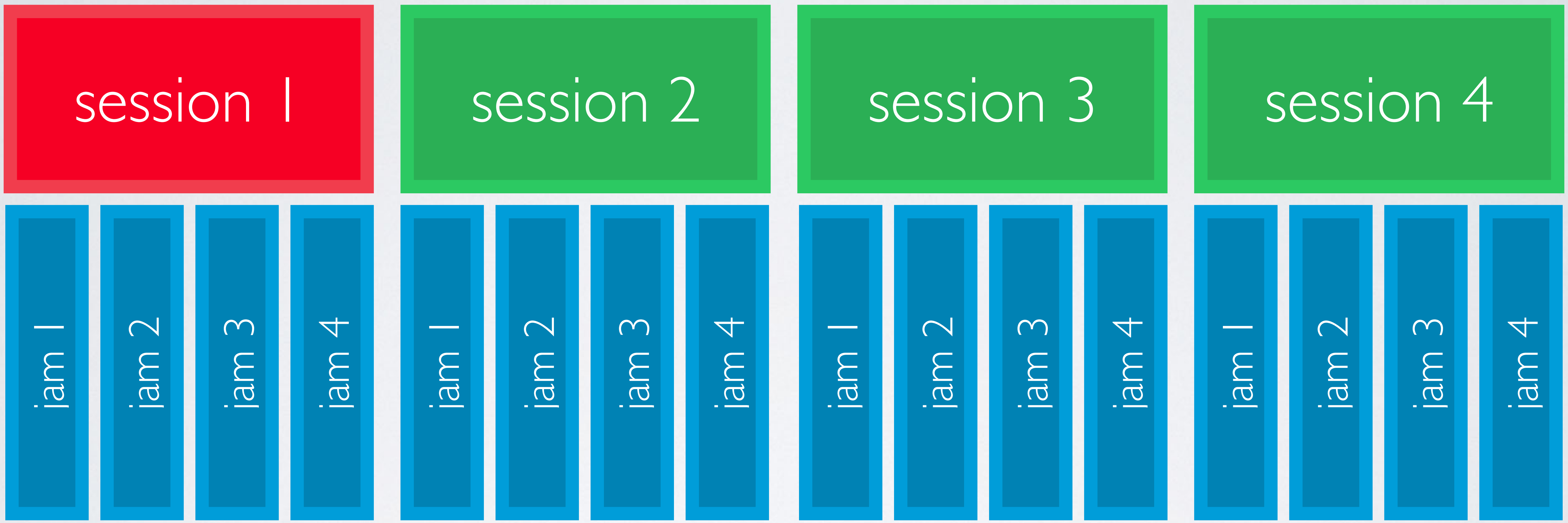
- not attaching numbers to things that aren't
- looking for *differences* in the data
- examining the *nature* of those differences


feature window (5 sec)
mode
touch activity
touch position
touch zone
accel
audio
musician

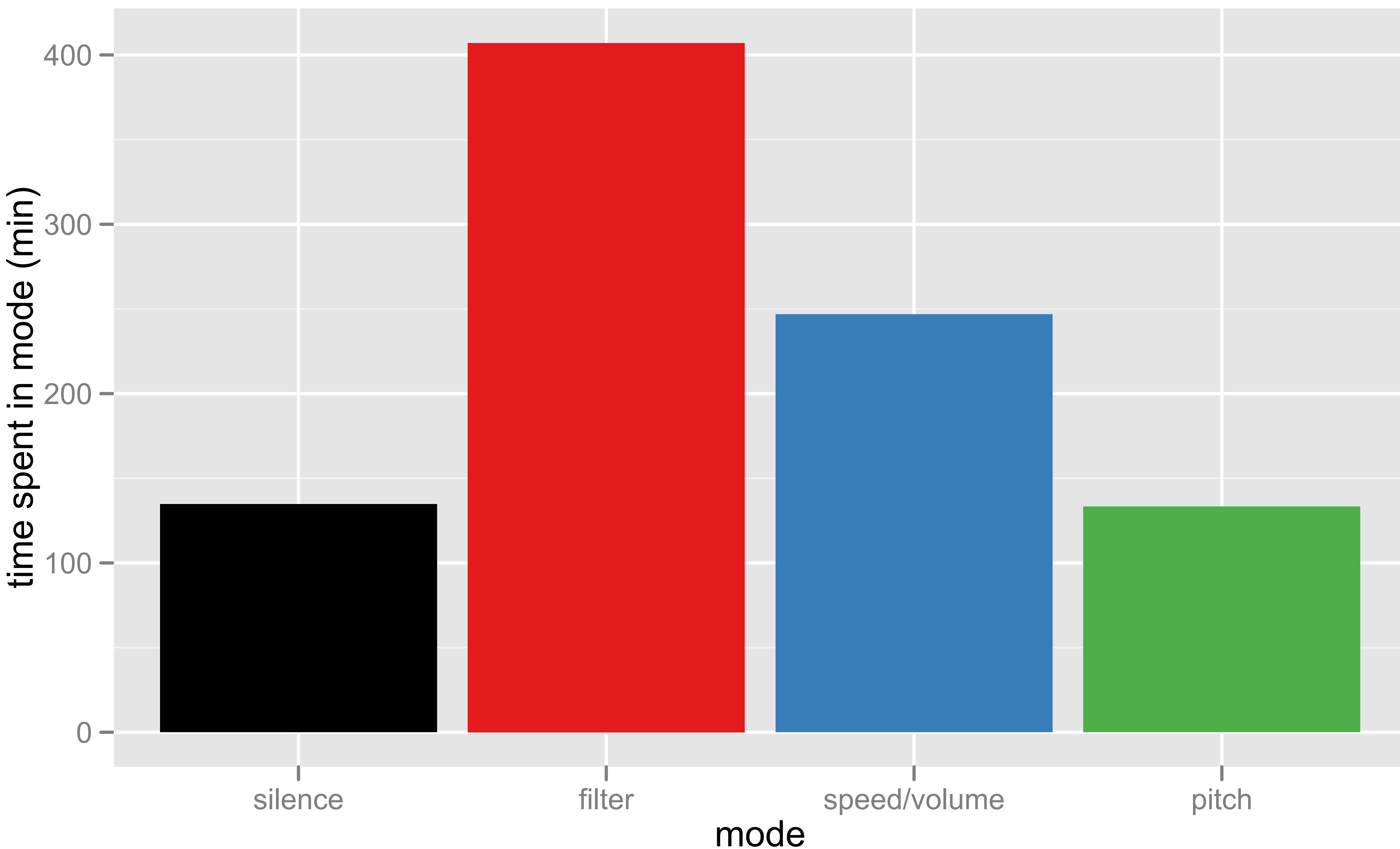


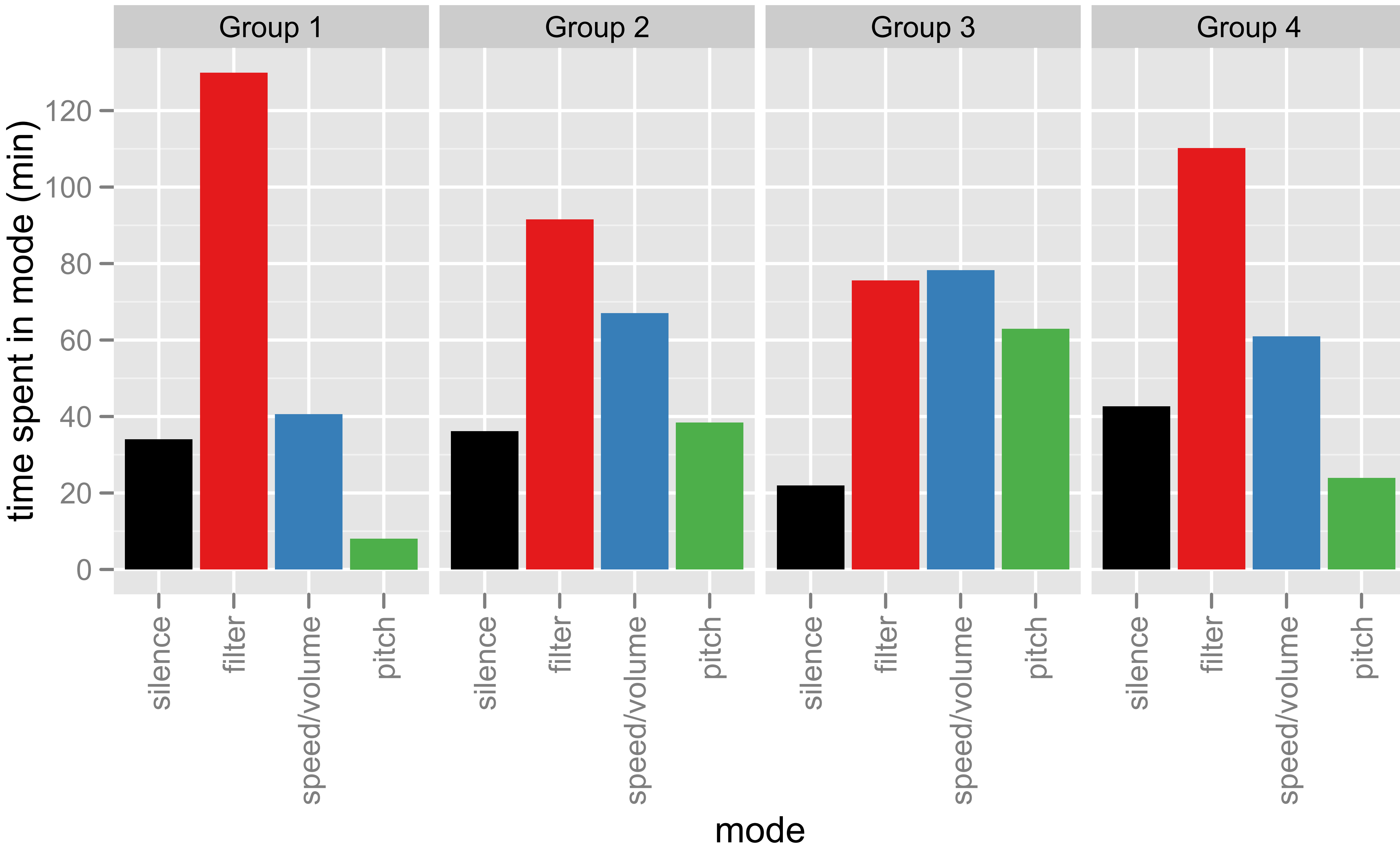
classification accuracy (vs chance)
0
1
2
3
4
5
6
session
jam
cross-validation partition
classifier
NB
SVM
RF


session 1
session 2
session 3
session 4
jam 1
jam 2
jam 3
jam 4
jam 1
jam 2
jam 3
jam 4
jam 1
jam 2
jam 3
jam 4
jam 1
jam 2
jam 3
jam 4

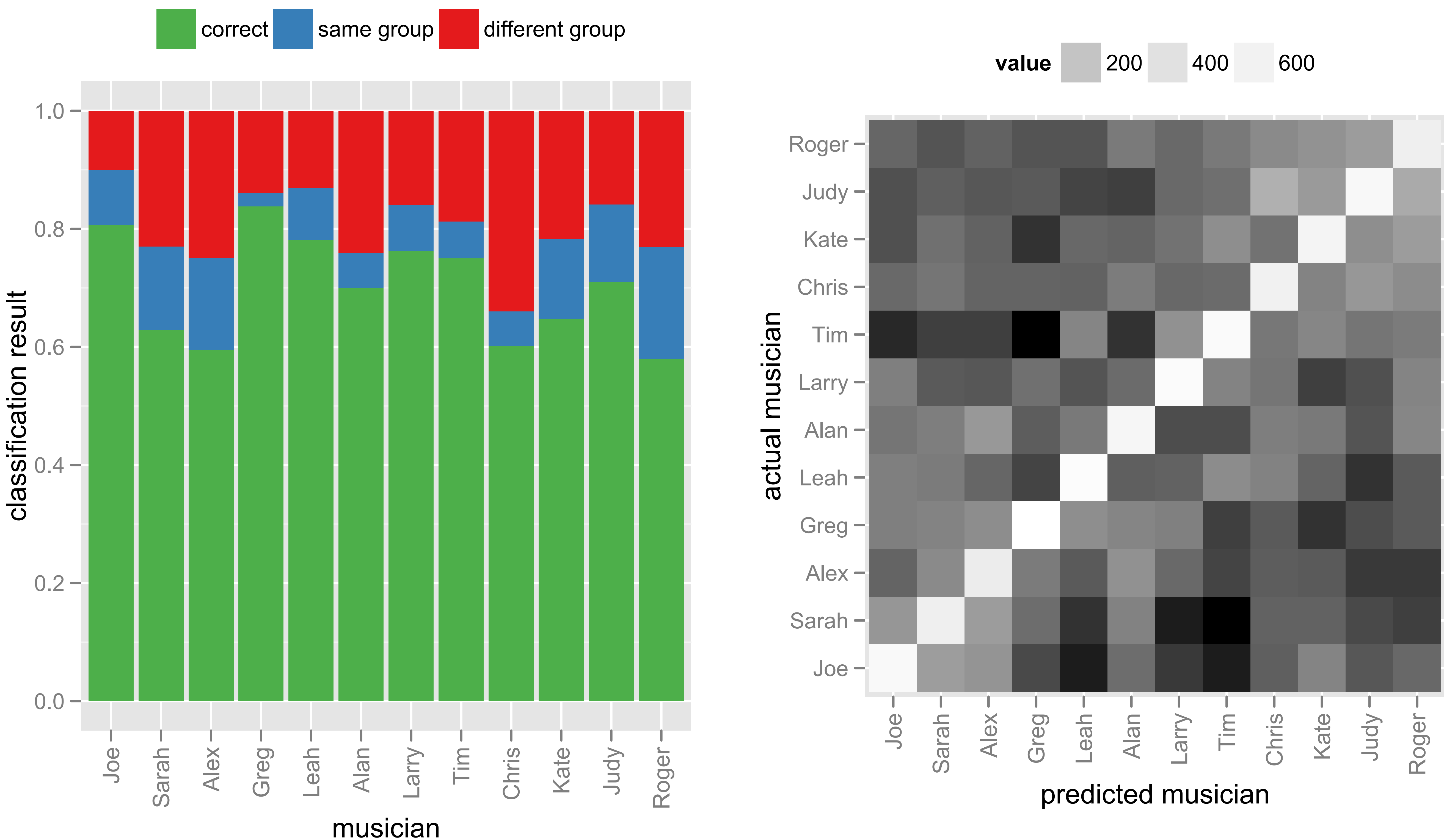

session 1
session 2
session 3
session 4
jam 1
jam 2
jam 3
jam 4
jam 1
jam 2
jam 3
jam 4
jam 1
jam 2
jam 3
jam 4
jam 1
jam 2
jam 3
jam 4


session 1
session 2
session 3
session 4
jam 1
jam 2
jam 3
jam 4
jam 1
jam 2
jam 3
jam 4
jam 1
jam 2
jam 3
jam 4
jam 1
jam 2
jam 3
jam 4


time spent in mode (min)
400
300
200
100
0
silence
filter
speed/volume
pitch
mode



Group 1
Group 2
Group 3
Group 4
time spent in mode (min)
0
20
40
60
80
100
120
silence
filter
speed/volume
pitch
mode



correct
same group
different group
classification result
1.0
0.8
0.6
0.4
0.2
0.0
Joe
Sarah
Alex
Greg
Leah
Alan
Larry
Tim
Chris
Kate
Judy
Roger
musician
value
200
400
600
actual musician
Roger
Judy
Kate
Chris
Tim
Larry
Alan
Leah
Greg
Alex
Sarah
Joe
predicted musician